



UNIVERSITY OF TOMORROW

# ASSIGNMENT - 1 [ USING AI TOOLS TO DEVELOP AND IMPROVE A C PROGRAM]

SUBJECT - PROGRAMMING IN C

10.09.2026

---

SUBMITTED BY

VAISHNAVI BAI PAI

COURSE :- B.TECH CSE (1ST YEAR)

SAP ID :- 5900036494 [ BATCH :- 34 ]

SUBMITTED TO

SUGANDHA SHARMA

## PART A {UNDERSTANDING THE CASE STUDY}

QUES.1 :- WHAT PROBLEM WAS RIYA TRYING TO SOLVE ?

ANS :-

Ans:- In the given case study Riya was trying to write a C program to arrange the marks of five different students. The program had to read the marks, computing the mean, highest, lowest marks of the provided students. Beside this, she was also counting the number of students those who were passed and the passing criteria was 40 marks or above.

QUES.2 :- WHY DID RIYA USE AN AI TOOLS ?

ANS :-

Ans:- According to the given information, Riya understood the basic concept of "for loop" and "array", even she was unable to write the C program according to provided information or requirements. That the reason she was used an AI tool to write the program.

QUES.3 :- WAS RIYA'S PROMPT CLEAR ? GIVE REASONS.

ANS:-

Ans:- Riya's prompt was clear because she specified the requirement of developing the C program around arrays and for loop for five students. However, the prompt did not ask for input validation, display whether the individual students passed and deal with invalid input.

## PART B {CODE UNDERSTANDING}

QUES.4:- WHAT IS THE USE OF THE ARRAY MARKS [5]?

ANS:-

Ans:- The array marks[5] stores the marks of the five students. The five elements, marks[0] to marks[4] corresponding to the marks of students 1 to student 5.

QUES.5:- WHY ARE HIGHEST AND LOWEST INITIALIZED WITH MARKS[0] ?

ANS :-

Ans:- The variables highest and lowest are initialized with marks[0] because marks[0] is the first value entered by the user. This allows them to compare the remaining value and update the variables accordingly.

QUES.6 :- EXPLAIN THE PURPOSE OF THIS CODE :

```
IF (MARKS[i] >= 40) {  
    PASSCOUNT++;}
```

ANS :-

Ans:- This statement checks whether the marks of the current student are equal to or more than 40. If true, it increments the passcount variable because the student has passed.

QUES.7 :- WHAT WILL BE THE MARKS ARE : **78 35 90 42 60**

ANS:-

Sol:-

- Sum =  $78 + 35 + 90 + 42 + 60 = 305$
- Average marks =  $\frac{305.0}{5} = 61.00$
- Highest marks = 90
- Lowest marks = 35
- Total number of passed students = 4



## PART C {AI CODE VERIFICATION}

QUES.8:- IS THE AI-GENERATED CODE IS CORRECT?

ANS:-

Ans:- The Code generated by the AI is correct for valid marks but does not handle invalid input. It stores the entered five marks in an array and calculates the average, highest, and lowest marks. It also counts the number of students who have secured more than 40 marks. However, the code does not reject marks less than zero and greater than 100 as invalid input.

QUES.9:- WHAT IMPROVEMENT CAN BE MADE IN THE AI-GENERATED CODE ?

ANS:-

Ans I would recommend improving the AI code by adding input validation so that only valid marks between 0 and 100 are stored. Other recommendations include validating the input from the scanf() function, using a constant for the number of students, and displaying whether the individual students have passed or failed.

QUES.10:- WHY IS IT IMPORTANT TO TEST AI-GENERATED CODE ?

ANS-

Ans:- The AI Code may contain errors such as syntax errors, incorrect conditional statements, missing requirements, and special case errors. Testing ensures that the code complies, passes all test cases, handles special cases, and meets all the requirements mentioned in the prompt. Testing also ensures that the code is well-written, follows coding standards, and is easy to maintain and modify.

## PART D { IMPROVED PROGRAM }

QUES : - MODIFY THE AI-GENERATED PROGRAM SO THAT IT ACCEPTS ONLY VALID MARKS BETWEEN 0 AND 100 . WRITE THE IMPROVEMENT CODE.

ANS:-

Sol:- The following improved program accepts marks only from 0 to 100. It also checks that the input is a number and prints each student's pass or fail result.

## PART E { PROMPT IMPROVEMENT }

QUES.11:- WRITE A BETTER PROMPT THAT RIYA COULD GIVE TO THE AI TOOL.

ANS :-

Ans. write a beginner-friendly C program to manage the marks of 5 students. Store the marks in an integer array and use for loops for input and processing. Only valid marks between 0 and 100 should be accepted invalid marks should be rejected. The program should display the class average with two decimal places, highest and lowest marks and a count of students who passed, where passing marks are 40 or above. The program should use simple variable names, proper indentation, comments, and explain the program step by step for a first-year B.Tech CSE students.

QUES.12:- WHY IS IMPROVED PROMPT BETTER ?

ANS:-

Ans:- The prompt is better because it specifies every detail about the desired program. It requests the use of for loop, input validation of percentage between 0 and 100, and calculation of average with two decimal places. It also asks for a comment and step by step explanation for a first-year B.Tech CSE student. This way the AI tool knows exactly what to do.

## PART F { OUTPUT TRACING }

QUES.13:- TRACE THE PROGRAM FOR THE FOLLOWING INPUT :

ANS:-



## PART 6 {CRITICAL THINKING QUESTIONS}

QUES.14:-SHOULD RIYA SUBMIT THE AI-GENERATED CODE DIRECTLY? WHY OR WHY NOT?

ANS:-

Ans:- No, Riya should not submit the AI Code directly to the person. She should review and understand the code before compiling and testing it. The original code lacks input validating, so she should improve it. Additionally, this way, she can learn the C-program.

QUES.15:- WHAT ARE TWO POSSIBLE MISTAKES AI COULD MAKE WHILE GENERATING THIS PROGRAM?

ANS:-

Sol:- The AI might have failed to restrict the input to marks between 0 and 100 or might have printed failed students for those with exactly 40 marks. It could have also made error in the loop condition or failed to calculate the average correctly.

QUES.16:- HOW DOES THIS CASE STUDY SHOW RESPONSIBLE USE OF AI ?

ANS:-

Ans:- Riya uses AI as a learning assistant, not as a replacement for her own work. She reviews the answer, finds the missing validation, improves the prompt, changes the code, and tests it with different inputs. This shows that AI output must be understood, verified and improved by the student.

## PART H {STUDENT ACTIVITY}

### ACTIVITY

## PART I {REFLECTION WRITING}

QUES.: - WRITE A SHORT PARAGRAPH OF 150-200 WORDS ON :

**"HOW AI HELPED RIYA LEARN C PROGRAMMING BETTER."**

ANSWER:

AI helped Riya learn C programming better by giving her a clear example of how different programming concepts work together in one program. She already knew basic array and for loop, but she was confused about using them for a complete marks-management problem. The AI-generated program showed her how to store the marks of five students in an array, take input using a loop, calculate the total and average, and find the highest and lowest marks using comparison statements. It also helped her understand how an if condition can count students who pass and marks of 40 or above. However, AI did not replace Riya's own learning. She checked the generated program and found that it did not validate marks between 0 and 100. She then improved the prompt, added validation and tested the program with different inputs. This taught her that AI can provide useful guidance and examples, but student must still understand, verify, test, and improve code themselves. Responsible use of AI means using it to support learning rather than blindly copying its output.







VAISHNAVI BAJPAI

590036494